Relaxation and Odd-Even Effect in Fermi Liquids

Kittipat Pongarunotai, Puripat Thumbanthu

Outline

1 Introduction

- k-space
- Fermi surface

2 Boltzmann equation

- Linearized collision operator

3 Odd-Even effect

1 Introduction

1.1 k-space

k-space

- Analogous to momentum space
- Each point $\mathbf{k}$ in k-space corresponds to a wave $\mathrm{e}^{\mathrm{i}\mathbf{k}}$
- **Fourier transform** converts between k-space to real space

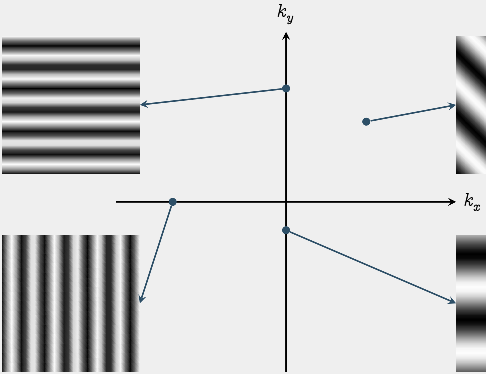

Figure 1: REPRESENTATIVE WAVES OF k-SPACE

- Distance from origin $\propto$ Wave frequency
- Direction on k-space $\implies$ Wave direction

1.2 Fermi surface

2 Boltzmann equation

- Dilute gas
- Only short range interactions

$$\frac{\partial f}{\partial t} + \mathbf{v} \cdot \nabla_{\mathbf{r}} f + \mathbf{F} \cdot \nabla_{\mathbf{v}} f = \left(\frac{\partial f}{\partial t} \right)_{\text{coll}} \tag{1}$$

$$\left(\frac{\partial f}{\partial t}\right)_{\text{coll}} \approx \int \dots \int \mathrm{d}^3\mathbf{p}_2 \, \mathrm{d}^3\mathbf{p}_1' \, \mathrm{d}^3\mathbf{p}_2' W(\mathbf{p}_1', \mathbf{p}_2' | \mathbf{p}, \mathbf{p}_2)[f_2(\mathbf{r}, \mathbf{r}, \mathbf{p}_1', \mathbf{p}_2') - f_2(\mathbf{r}, \mathbf{r}, \mathbf{p}_1, \mathbf{p}_2)]$$

2

In equilibrium, $\left(\frac{\partial f}{\partial t}\right)_{\text{coll}} = 0$

2.1 Linearized collision operator

- No chemical potential

3 Odd-Even effect